

Yahya SALEH

CURRICULUM VITAE

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Research interests. Approximation theory, scientific machine learning, statistical learning theory, computational quantum molecular physics.

EMPLOYMENT

since 04/2025	Consultant, d-fine GmbH
04/2023 – 03/2025	Postdoctoral Researcher, Department of Mathematics, Universität Hamburg
04/2020 – 03/2023	Doctoral Researcher, Department of Mathematics, Universität Hamburg and Center for Free-Electron Laser Science (CFEL), DESY
05/2018 – 09/2018	Research Assistant, Center for Free-Electron Laser Science (CFEL), DESY

EDUCATION

04/2020 – 05/2023	Dr. rer. nat. in Mathematics (<i>magna cum laude</i>), Universität Hamburg Graduate school: Data Science in Hamburg – Helmholtz Graduate School for the Structure of Matter (DASHH) Thesis: <i>Spectral and active learning for enhanced and computationally scalable quantum molecular dynamics</i> Supervisors: Prof. Dr. Armin Iske, Prof. Dr. Jochen Küpper, Dr. Andrey Yachmenev
2017 – 10/2019	M.Sc. in Mathematical Modelling in Engineering (MathMods) , joint Erasmus Mundus degree , Università degli Studi dell'Aquila, Universität Hamburg, Universitat Autònoma de Barcelona Thesis: <i>On non-adiabatic dynamics of deflected molecular beams</i> Supervisors: Prof. Dr. Armin Iske, Prof. Dr. Jochen Küpper, Dr. Andrey Yachmenev
2012 – 09/2017	B.Eng. in Electrical Engineering (Electronics and Telecommunications) , Damascus University Thesis: <i>Designing a cellular-based tracking system using machine learning and Kalman filtering</i> Supervisors: Prof. Dr. Hiyam Khaddam

TEACHING EXPERIENCE

Universität Hamburg, Department of Mathematics

Winter 2024/25	<i>Software Practicum</i> (M.Sc.) — Lecturer, lectures and exercise sessions, in German
Summer 2024	<i>Mathematical Machine Learning</i> (M.Sc.) — Lecturer, lectures and exercise sessions, in English
Winter 2023/24	<i>Higher Analysis</i> (B.Sc.) — Teaching assistant to Prof. Dr. Armin Iske, exercise sessions, in German
Summer 2023	<i>Numerical Methods for Partial Differential Equations</i> (M.Sc.) — Teaching assistant to Prof. Dr. Hendrik Ranocha, exercise sessions, in English

STUDENTS MENTORED

2024–2025	Sharlet Shaji, Master's student, Universität Hamburg, Department of Mathematics — Spectral learning for solving the time-dependent Schrödinger equation; co-supervised with Prof. Dr. Armin Iske
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RESEARCH STAYS AND GRANTS

08/2024 – 09/2024	Research stay with Prof. Dr. Martin Frank, Karlsruhe Institute of Technology (KIT), funded by a research-stay grant of the Helmholtz Information and Data Science Academy (HIDA)
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PUBLICATIONS

Reverse chronological order. Key publications are marked with *.

Peer-reviewed articles

- [1] E. Vogt, Á. F. Corral, **Y. Saleh**, “Adaptive Vibrational Coordinates via Symmetry-Aware Normalizing Flows,” *Journal of Chemical Theory and Computation* **22**(12), 6137 (2026). doi:10.1021/acs.jctc.6c00224
- * [2] **Y. Saleh**, A. Iske, “Inducing Riesz Bases in L^2 via Composition Operators,” *Complex Analysis and Operator Theory* **20**, 21 (2026). doi:10.1007/s11785-025-01874-5
- [3] A. Yachmenev, E. Vogt, Á. F. Corral, **Y. Saleh**, “Taylor-mode automatic differentiation for constructing molecular rovibrational Hamiltonian operators,” *The Journal of Chemical Physics* **163**(7), 072501 (2025). doi:10.1063/5.0287347
- [4] E. Vogt, Á. F. Corral, **Y. Saleh**, A. Yachmenev, “Transferability and interpretability of vibrational normalizing-flow coordinates,” *The Journal of Chemical Physics* **163**(15), 154106 (2025). doi:10.1063/5.0285954
- * [5] **Y. Saleh**, Á. F. Corral, E. Vogt, A. Iske, J. Küpper, A. Yachmenev, “Computing Excited States of Molecules Using Normalizing Flows,” *Journal of Chemical Theory and Computation* **21**(10), 5221–5229 (2025). doi:10.1021/acs.jctc.5c00590
- * [6] V. Menden, **Y. Saleh**, A. Iske, “Bounds on the Generalization Error in Active Learning,” *Proceedings of the 6th Northern Lights Deep Learning Conference (NLDL)* **265**, 168–175 (2025). proceedings.mlr.press/v265/menden25a.html
- [7] T. Wenzel, **Y. Saleh**, A. Iske, “Two-Layered and Deep Kernels as Data-Adapted Kernels,” *DEEPK 2024: International Workshop on Deep Learning and Kernel Machines* (2024). www.esat.kuleuven.be/stadius/E/DEEPK2024/7_two_layered_and_deep_kernels_a.pdf
- [8] **Y. Saleh**, A. Iske, A. Yachmenev, J. Küpper, “Augmenting Basis Sets by Normalizing Flows,” *Proceedings in Applied Mathematics and Mechanics* **23**(1), e202200239 (2023). doi:10.1002/pamm.202200239
- * [9] **Y. Saleh**, V. Sanjay, A. Iske, A. Yachmenev, J. Küpper, “Active learning of potential-energy surfaces of weakly bound complexes with regression-tree ensembles,” *The Journal of Chemical Physics* **155**(14), 144109 (2021). doi:10.1063/5.0057051

Preprints

- * [10] **Y. Saleh**, “Convergence theory for Hermite approximations under adaptive coordinate transformations,” arXiv:2604.16975 [math.NA] (2026).
- [11] Á. F. Corral, **Y. Saleh**, “Enhancing polynomial approximation of continuous functions by composition with homeomorphisms,” arXiv:2512.13740 [math.NA] (2025).

CONTRIBUTIONS TO INTERNATIONAL CONFERENCES AND WORKSHOPS

Invited talks

2026	“Learning, sharing, and using optimized vibrational coordinates,” ExoMol 2026, University College London, United Kingdom
2025	“Normalizing flows: from learning probability distributions to basis-set discovery,” Linköping University, Sweden

2024	"Learning basis sets in L^2 and application to computing excited states of molecules," BIRS Workshop on Modern Methods for Differential Equations of Quantum Mechanics , Banff, Canada
2023	"Flow-induced bases and application to quantum molecular physics," Machine Learning in Engineering, TU Hamburg, Germany
2023	"Augmenting bases with normalizing flows for solving Schrödinger equations," DPG SAMOP Spring Meeting, Germany
2021	"Applications of machine learning in quantum simulations of hydrogen bond dynamics," Machine Learning for Quantum X , online
<i>Contributed talks</i>	
2026	"Learning bases via normalizing flows, applications to solving molecular Schrödinger equations and image compression," International Conference on Scientific Computing and Machine Learning, Bath, United Kingdom
2025	"Learning basis sets using unitary and bounded bijective operators," International Conference in Numerical Mathematics and Scientific Computing, Uppsala University, Sweden
2024	"Spectral learning for solving molecular Schrödinger equations," PINN-PAD: Physics-Informed Neural Networks in Padova , Padova, Italy
2024	"Augmenting spectral methods with invertible neural networks and application to quantum molecular physics," ALGORITMY 2024 , Podbanské, Slovakia
2022	"Spectral learning for solving the Schrödinger equation for molecules," Computational Methods in Applied Mathematics (CMAM), TU Wien, Austria
2022	"Deep spectral methods for solving variational problems arising from differential equations," GAMM Annual Meeting, RWTH Aachen, Germany
2022	"Deep spectral methods for solving variational problems arising from differential equations," HCM Workshop: Synergies Between Data Science and PDE Analysis, Universität Bonn, Germany
2022	"Spectral learning for (ro-)vibrational calculations of weakly bound molecules," DPG SAMOP Spring Meeting, Germany
2021	"Active learning for computing potential-energy surfaces of molecules," Bunsentagung, online
<i>Posters</i>	
2025	"Bounds on the generalization error in active learning," 6th Northern Lights Deep Learning Conference (NLDL), Tromsø, Norway
2024	"Computing excited states of molecules using normalizing flows," High-Dimensional Quantum Dynamics Conference, Hamburg, Germany
2022	"Deep learning for computing spectra of molecules," Opening Symposium, Center for Data and Computing in Natural Sciences (CDCS), Hamburg, Germany
2022	"Deep learning for computing rovibrational spectra of molecules," European CFEL and DESY Photon Science Users' Meeting, Hamburg, Germany
2021	"Applications of machine learning in quantum simulations of hydrogen bond dynamics," Warsaw Summer School for Quantum Physics and Chemistry, University of Warsaw, Poland
<i>Schools and hackathons attended</i>	
2022	Hausdorff School: Foundational Methods in Machine Learning, Universität Bonn, Germany
2021	Helmholtz H3 Hackathon

LANGUAGES AND SKILLS

Languages	Arabic (native) · English (C2) · German (C2) · Italian (C1) · French (B2)
Programming	Python (JAX, NumPy, SciPy), Julia, C++, MATLAB, Java, Bash
Tools	Git, GitHub/GitLab, CI/CD (Jenkins), Docker, Kubernetes